power transfer-based flux generation to create a magnetic field inside an electric motor. This enables us to achieve performance comparable to or better than magnet-based motors while eliminating dependence on rare-earth magnets," Manish explains.

Vimag Labs claims a system-level cycle efficiency of 92-93%, matching conventional permanent-magnet setups. "Depending on the motor size and application, motor efficiency can reach approximately 93-96%, with around 5-7% higher torque density than comparable permanent magnet synchronous motor (PMSM) technology, enabling smaller, lighter motors for a given torque output," says Manish.

The startup holds five approved patents, including the foundational filing for its virtual magnet technology, with at least eight additional patents in the pipeline across India, Europe, North America, and other global markets.

While the startup is currently in the early commercialisation stage, it is already generating revenue. Vimag Labs has supplied initial samples and small component quantities, monetising its technology through direct product sales as well as customer-funded non-recurring engineering (NRE) projects.

The company expects to begin deploying certain control systems within the next few months and aims to start production of complete virtual magnet systems before the end of 2026.

"We are initially focusing on two- and three-wheelers, followed by electric buses and trucks, while passenger-car programmes are being developed in parallel. Some two- and three-wheeler programmes are already approaching the commercial stage, while passenger-car projects are currently being developed through paid proof-of-concept programmes", explains Manish.



Manish Seth, Founder, Vimag Labs

Manish shares that the startup has around 22 employees globally, covering India, Europe, and North America. At the Bengaluru facility, the company has an R&D, prototyping, and testing centre. A production facility is currently being established in North India. The startup follows a hybrid manufacturing model, keeping core activities such as electronics assembly, software, integration, testing, and deployment in-house while outsourcing commodity components such as machining.

For product development, Vimag Labs follows a comprehensive testing cycle involving digital simulations, digital twins, prototypes, dynamometer testing, hardware-in-the-loop testing, pilot vehicles, and customer road testing. Feedback from each stage is incorporated into subsequent design and software improvements.

On the challenges faced, Manish says, "There were no established tools for designing and simulating this type of motor. We had to begin with calculations and develop our own software and systems to accelerate the design process. The technology requires the motor, electronics, software, and firmware to work together reliably and deliver the required performance. We therefore had to develop and validate several elements simultaneously."

Decoding the name, Manish explains that 'Vimag' stands for 'Virtual Magnet', with 'VI' representing Virtual and 'MAG' representing Magnet, while 'Labs' underscores the company's ownership and ongoing development of the core technology.

The startup is also developing its own design and simulation tools to optimise magnet-free motor performance. "Since the technology requires coordinated hardware, electronics, software, firmware, and automotive integration, we have built a multidisciplinary engineering team covering these areas," highlights Manish.

Looking ahead, Vimag Labs plans to expand into higher-power applications and new industry segments. In the near term, the company remains focused on fulfilling existing customer orders and further maturing its technology before aggressively scaling manufacturing capacity.

—Nitisha Dubey

3 EVE NET TURNS SPARE SMARTPHONE CAPACITY INTO AN AI CLOUD

What if the phone in your pocket could help run AI? A Mumbai-based startup wants to cut AI computing costs by tapping into millions of smartphones.

Jash Jetly, co-founder and CEO of EVE NET, points out that when Google started in 1999, storing data was expensive, much like artificial intelligence (AI) computing is today. Yet Google made web search free. Imagine if it had charged one dollar for every search—most people would never have used it. That made Jash Jetly question why AI should be any different. AI is trained on public